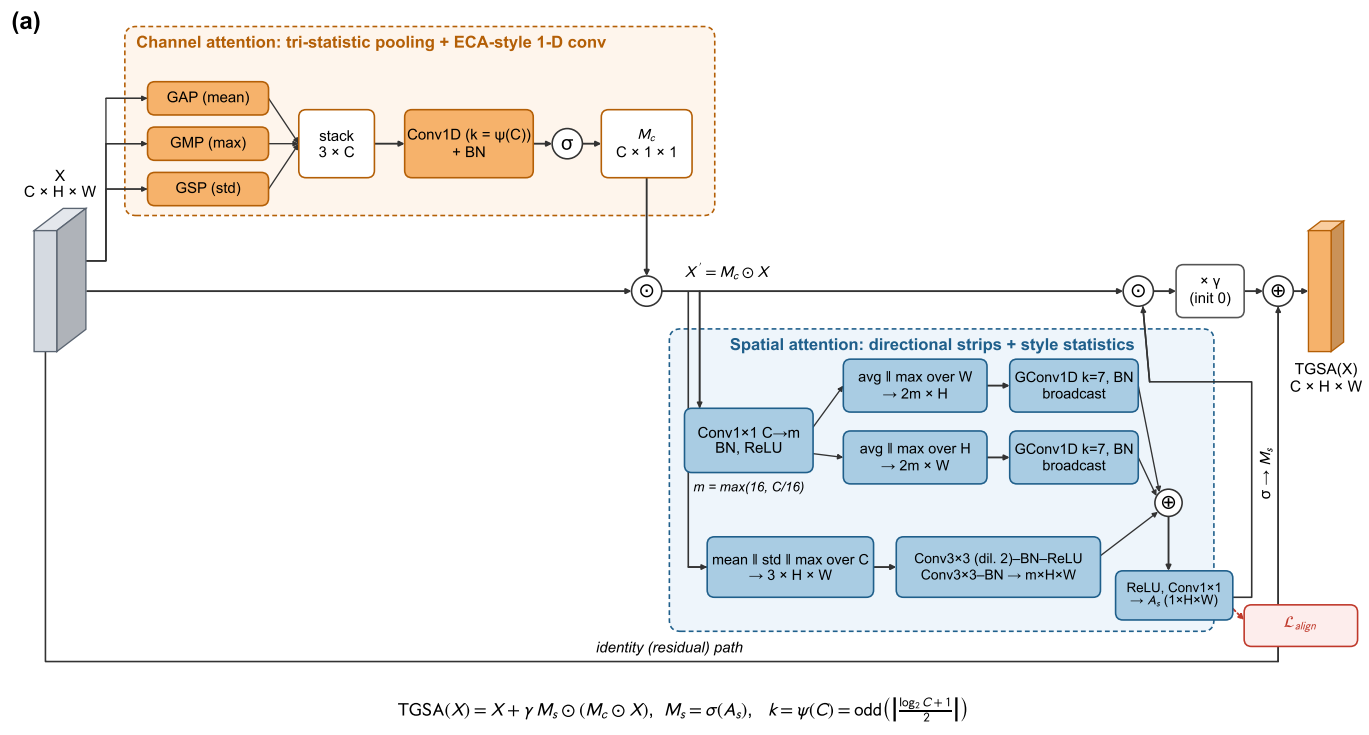


$$\mathrm{TGSA}(X) = X + \gamma\, M_s \odot (M_c \odot X), \quad M_s = \sigma(A_s), \quad k = \psi(C) = \mathrm{odd}\left(\left|\frac{\log_2 C + 1}{2}\right|\right)$$

(b) Learned TGSA spatial attention $M_s = \sigma(A_s)$ of the trained ResNet50 TG-SSPANet on held-out test slices (P2 fold 1; green: ground-truth tumour)

